# SML Assignment-3

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# 1 Question 1: Ridge and Lasso Regression

## 1.1 Preprocessing

The MNIST dataset was used, restricted to classes 0, 1, and 2. The images were flattened and normalized to the range $[0, 1]$. Principal Component Analysis (PCA) was applied to reduce the dimensionality to $p = 10$.

For multi-class classification, one-hot encoding was used:

$$y_k = \begin{cases} 1 & \text{if sample belongs to class } k \\ 0 & \text{otherwise} \end{cases}$$

## 1.2 Ridge and Lasso Regression

Ridge regression minimizes:

$$||Y - Xw||^2 + \lambda ||w||^2$$

Lasso regression minimizes:

$$||Y - Xw||^2 + \lambda ||w||_1$$

Models were trained for:

$$\lambda \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10, 100\}$$

### 1.3 MSE vs $\lambda$

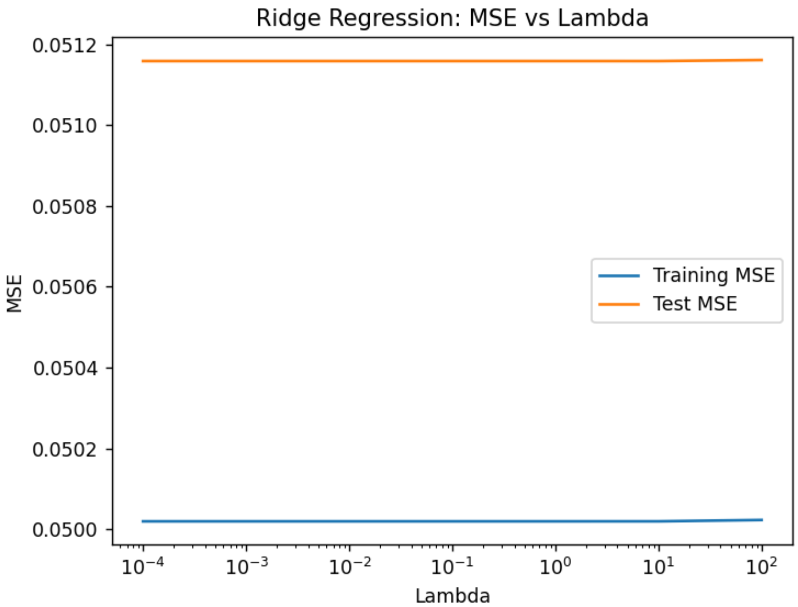


Figure 1: Ridge Regression: Training and Test MSE vs $\lambda$

**Observation:** As $\lambda$ increases, the model becomes more regularized. Training error increases while test error initially decreases and then increases, showing the bias-variance tradeoff.

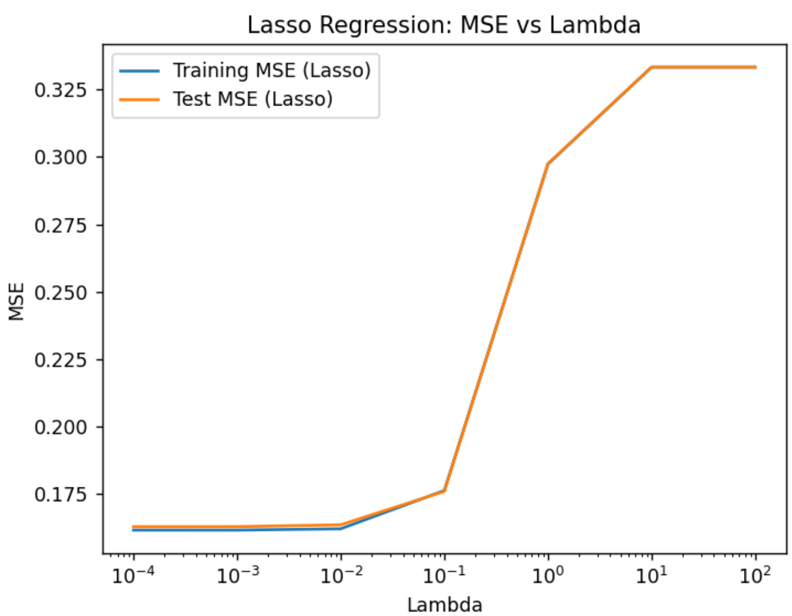


Figure 2: Lasso Regression: Training and Test MSE vs $\lambda$

**Observation:** Lasso shows a similar trend but enforces sparsity, leading to slightly higher bias for large $\lambda$.

### 1.4 Lasso Sparsity

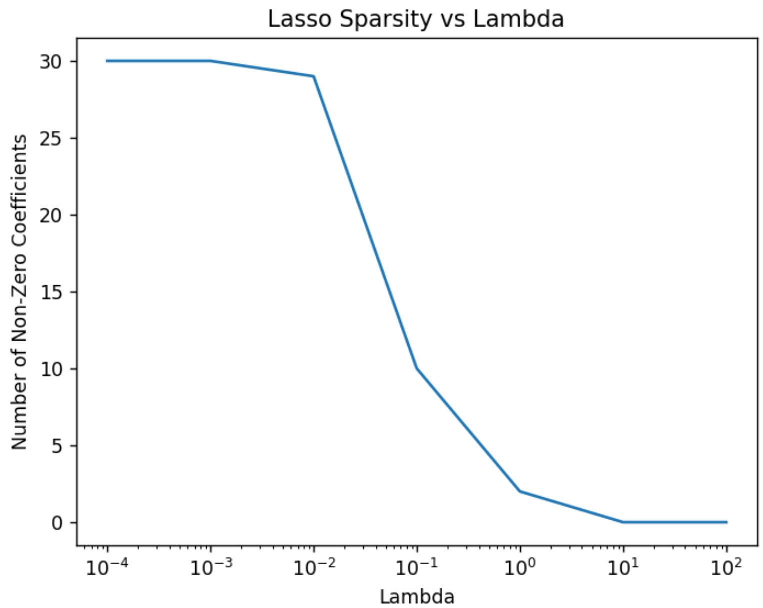


Figure 3: Number of Non-Zero Coefficients vs $\lambda$

**Observation:** As $\lambda$ increases, the number of non-zero coefficients decreases, demonstrating feature selection behavior.

### 1.5 Regularization Path

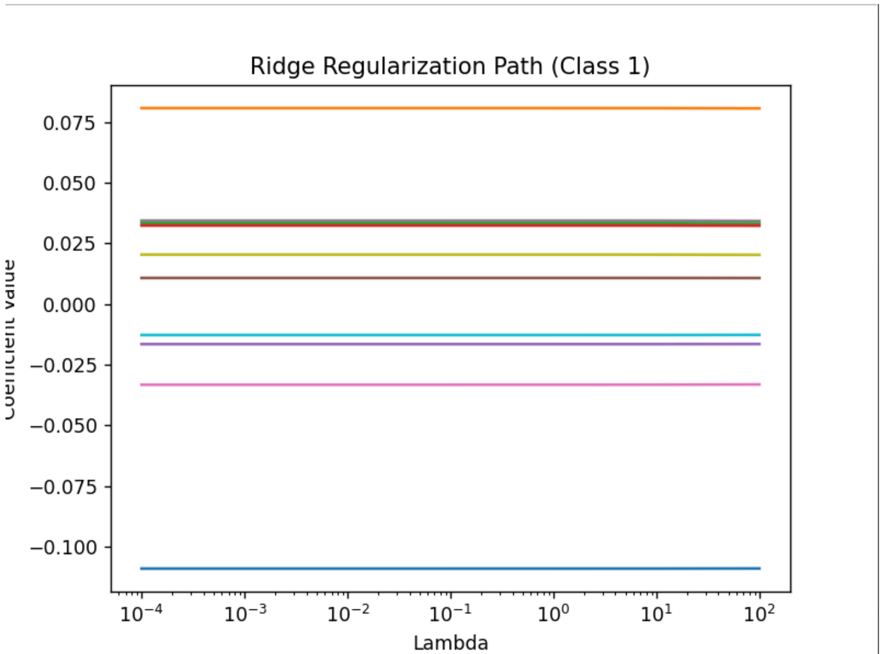


Figure 4: Ridge Regularization Path (Class 1)

**Observation:** Coefficients shrink smoothly as $\lambda$ increases.

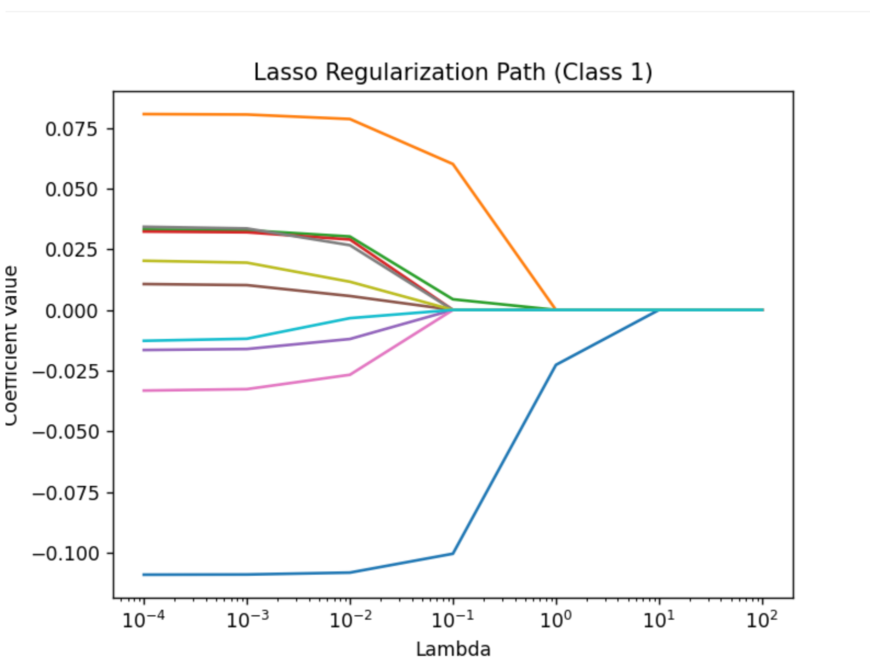


Figure 5: Lasso Regularization Path (Class 1)

**Observation:** Many coefficients become exactly zero as $\lambda$ increases.

### 1.6 Model Complexity vs PCA Dimension

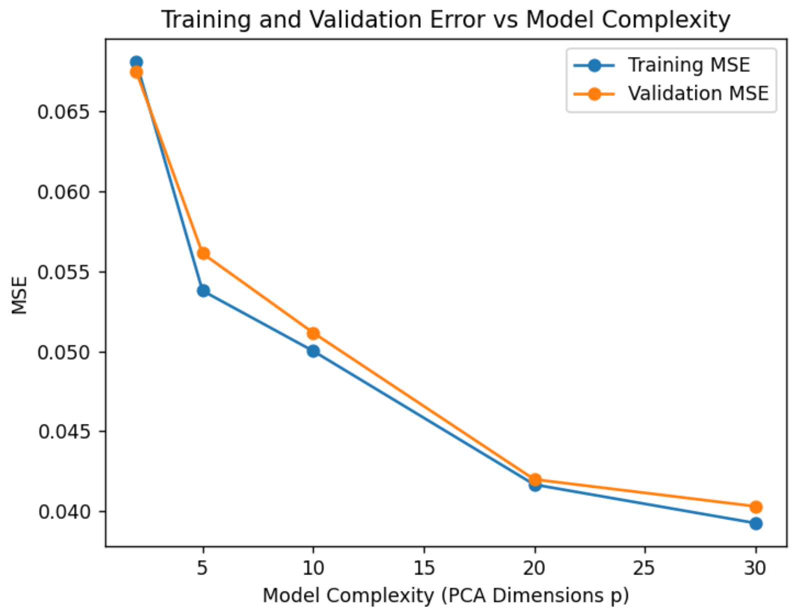


Figure 6: Training and Test Error vs PCA Dimension

**Observation:** Increasing PCA dimensions initially improves performance, but beyond a point, the error stabilizes, indicating that most useful variance is captured in fewer components.

### 1.7 Final Results

- Best Ridge $\lambda$: (10)
- Best Lasso $\lambda$: (0.0001)
- Ridge Test Accuracy: (0.9605973943438195)
- Lasso Test Accuracy: (0.9637750238322211)

**Conclusion:** Ridge provides stable performance, while Lasso performs feature selection. Both models achieve comparable classification accuracy.

## 2 Question 2: Decision Trees, Bagging and Random Forest

### 2.1 Methodology

The dataset was preprocessed similarly and reduced to $p = 10$ using PCA.

A decision tree with 3 terminal nodes was constructed using the Gini index:

$$Gini(S) = 1 - \sum_k p_k^2$$

The split minimizing weighted Gini was selected:

$$Gini_{weighted} = \sum_i \frac{|S_i|}{|S|} Gini(S_i)$$

**Threshold Choice:** The median was used as the split threshold, as it is robust to outliers and ensures balanced partitions.

### 2.2 Results: Single Tree

- Accuracy: 0.7817
- Class 0 Accuracy: 0.7755
- Class 1 Accuracy: 0.9973
- Class 2 Accuracy: 0.5504

**Observation:** The model performs very well on class 1 but struggles with class 2, indicating class imbalance or overlapping features.

### 2.3 Bagging

Five bootstrap datasets were created and trees were trained.

- Average OOB Error: 0.2159
- Accuracy: 0.7766

**Observation:** Bagging slightly stabilizes the model but does not significantly improve accuracy due to the simplicity of individual trees.

### 2.4 Random Forest

Random feature selection was introduced with $k = 3$.

- OOB Error: 0.2621
- Accuracy: 0.7598

**Observation:** Random Forest performs slightly worse in this case due to limited tree depth and small number of trees.

**Conclusion:** While Random Forest typically improves performance, here the shallow trees and small ensemble size limit its effectiveness.

# 3 Question 3: Decision Stump and Bagging (Regression)

## 3.1 Methodology

Fashion-MNIST dataset was used with classes 0, 1, and 2. Preprocessing included flattening, normalization, and PCA with $p = 10$.

A decision stump was trained by minimizing Sum of Squared Residuals:

$$SSR = \sum_{i \in L} (y_i - \bar{y}_L)^2 + \sum_{i \in R} (y_i - \bar{y}_R)^2$$

Candidate thresholds were midpoints between sorted values.

## 3.2 Results

- Single Stump MSE: (0.3629439763726561)
- Bagging Test MSE: (0.3589074283395048)
- Average OOB Error: (0.35320137333178625)

## 3.3 Comparison Plot

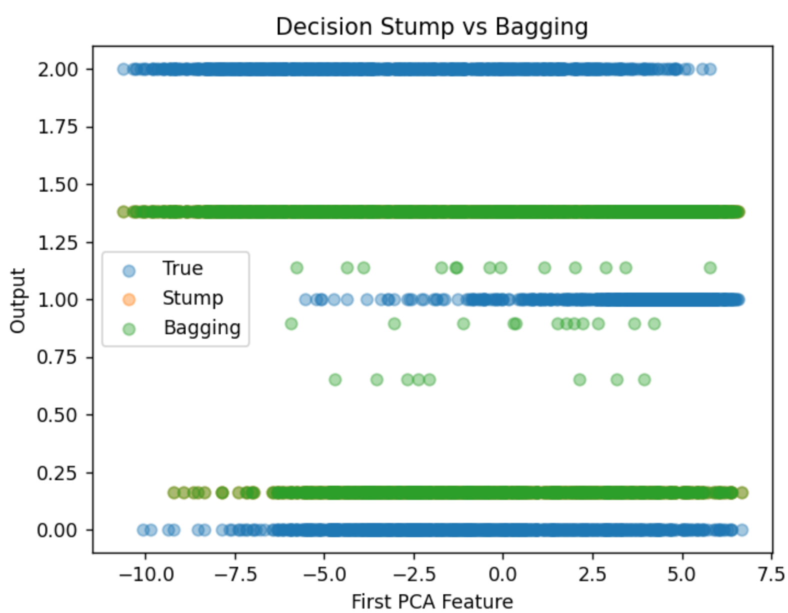


Figure 7: Decision Stump vs Bagging Predictions

**Observation:** The single stump produces a piecewise constant prediction, while bagging smooths the predictions by averaging multiple models, reducing variance.

### 3.4 Conclusion

Bagging significantly improves stability and reduces variance compared to a single decision stump, resulting in better generalization.

## 4 Overall Conclusion

This assignment demonstrates:

- PCA effectively reduces dimensionality without major performance loss
- Ridge and Lasso control overfitting via regularization
- Bagging reduces variance by averaging models
- Random Forest introduces feature randomness, but requires sufficient model complexity